Establishing a Baseline to Judge Error

When diagnosing bias and variance, simply looking at the error percentages (J_{train}, J_{cv}) isn't enough. The terms "high" and "low" are relative. To understand if an error is truly "high," you must compare it to a **baseline level of performance**.

What is a Baseline? 🤔

A **baseline** is the level of error you can reasonably hope your algorithm will eventually achieve. It represents the "best possible" or "unavoidable" error for your task.

Common ways to establish a baseline:

- **Human-Level Performance:** This is the best baseline for "unstructured data" problems where humans are skilled (e.g., speech recognition, computer vision, text understanding).
- **Competing Algorithm Performance:** The error rate of a previous system or a competitor's algorithm.
- **Prior Experience:** A reasonable guess based on past work.

A New Framework for Diagnosis

By introducing a baseline, we can more accurately diagnose bias and variance by looking at **two distinct gaps**:

1. **Gap 1: Bias (Avoidable Bias)**
    - **What to compare:** `(Baseline Error) <--> (Training Error, $J_{train}$)`
    - **Diagnosis:** If this gap is large, it means your algorithm isn't even matching the baseline performance on the data it trained on. This is a clear sign of **high bias (underfitting)**.
2. **Gap 2: Variance (Overfitting)**
    - **What to compare:** `(Training Error, $J_{train}$) <--> (Cross-Validation Error, $J_{cv}$)`
    - **Diagnosis:** If this gap is large, it means your algorithm is generalizing poorly to new data. This is a clear sign of **high variance (overfitting)**.

Examples in Practice (Speech Recognition)

Let's analyze three scenarios for a speech recognition task.

Scenario	Baseline (Human)	J_{train} (Training Error)	J_{cv} (CV Error)	Gap 1 (Bias)	Gap 2 (Variance)	Diagnosis
Initial (Flawed) Diagnosis	(Not Used)	10.8%	14.8%	-	-	*Looks like high bias?*
1. Correct Diagnosis	**10.6%**	**10.8%**	**14.8%**	**0.2% (Low)**	**4.0% (High)**	**High Variance**
2. High Bias Example	**10.6%**	**15.0%**	**15.1%**	**4.4% (High)**	**0.1% (Low)**	**High Bias**
3. High Bias & Variance	**10.6%**	**15.0%**	**19.7%**	**4.4% (High)**	**4.7% (High)**	**High Bias & High Variance**

Analysis of Scenario 1 (The Key Insight)

- **Flawed Diagnosis:** Without a baseline, a 10.8% training error *looks* high, leading you to incorrectly believe you have a high bias problem.
- **Correct Diagnosis:** Once you know the human-level error is 10.6%, you see your algorithm is only 0.2% worse than the baseline. The bias is actually **low**. The *real* problem is the large 4.0% gap between training and validation error, indicating **high variance**.